

University of Mumbai

S.E Second Year 2017 - 2018 June

BE Computer Engineering - Semester 4 (SE Second Year)

Database Management Systems

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- N.B. : (1) Question Number 1 is compulsory
(2) Solve any three Questions from the remaining.
(3) Make suitable assumptions if needed

1. (a) Construct an E-R diagram for a Library Management System. Convert the E-R Diagram to Tables. 10
 - (b) Explain Authorization in SQL. 5
 - (c) List four significant differences between file processing system and database management system 5
 2. (a) Explain Types of Integrity Constraints with example. 10
 - (b) Write SQL queries for the given database 10
- Employee(eid, emp-name, street, city)
- Works(eid, cid, salary)
- Company(cid, comp-name, city)
- Manager (eid, manager-name)
- (i) Find the names of all the employees having 'S' as first letter in their Names
 - (ii) Display the annual salary of all the employees.
 - (iii) Find the name, street and city of all employees who work for "Accenture" and earn more than 30,000.
 - (iv) Give total number of employees
3. (a) What is an attribute? Explain different types of attributes with examples. 10
 - (b) What is Normalization? Explain 1NF, 2NF, 3NF and BCNF. 10

4. (a) Explain following terms with examples 10
- (i) Weak Entity Set
 - (ii) Data Independence
 - (iii) Extended ER features
 - (iv) Total and Partial participation
- (b) Explain any five Relational Algebra Operators in detail. 10
- 5 (a) What is Transaction? Discuss the ACID properties of Transaction. 10
- (b) Describe the Overall architecture of DBMS with suitable Diagram. 10
- 6 (a) Explain log based recovery. 10
- (b) Write a note on 10
- 1) Armstrong axioms
 - 2) Thomas write rule
